

had clinically stable disease. 3 out of 4 patients with minimal changes or increased CNR ($m=0.7, 0.84, 0.94$ & 1.54) during treatment had clinical tumor progression with follow-up.

Conclusions: Preliminary analysis suggests a positive correlation between MRSI changes in tumor metabolic activity at 3 weeks of treatment and disease control. MRSI obtained during the course of RT provides valuable functional tumor information that could help to design patient-specific image-guided treatment interventions.

Author Disclosure: M. Muruganandham, None; J.E. Bayouth, Siemens Medical Solutions, B. Research Grant; C.M. Anderson, None; M.C. Smith, None; J.M. Buatti, None.

2106 Effect of Time Interval between Surgical Resection and Radiotherapy on Survival of Patients with Glioblastoma

M. Abacioglu, Z. Akgun, A. Ucuncu Kefeli, B. M. Atasoy, H. B. Caglar
Marmara University Hospital, Istanbul, Turkey

Purpose/Objective(s): The relationship between the delay in radiotherapy (RT) and outcome of patients with Glioblastoma (GBM) has been explored in a recent study analyzing the database of 16 RTOG studies (Blumenthal D, et al: J Clin Oncol 27: 733-9, 2009). This study suggests that, there is no evident reduction in survival by delaying initiation of RT and an unanticipated yet significantly superior outcome was identified for patients for whom RT was delayed beyond 4 weeks from surgery (Sx). Data from a study using rat models to examine brain Sx followed by radiation at differing onsets suggests that early initiation of radiation (within 1 to 2 weeks) after Sx, compared with 3 weeks or more, may result in higher levels of tissue damage (Peker S, et al: J Neuro-onc 70: 17-21, 2004). Decrease in radiosensitivity due to hypoxia after tissue injury may explain this unexpected outcome. We aimed to investigate the effect of time interval between surgical resection and RT on survival of GBM patients treated in a single institution.

Materials/Methods: A retrospective analysis was performed for 145 GBM patients who had undergone Sx and RT between 1997 and 2008. Patients were categorized into five groups on the basis of interval between Sx and start of RT; ≤ 2 weeks, 2-3 weeks, 3-4 weeks, 4-5 weeks, more than 5 weeks. Effect of time interval and other prognostic factors like age, KPS score, type of resection, gender, addition of temozolomide on overall survival were assessed by univariate and multivariate analysis.

Results: Median age was 53 years (range 22-78 years). Female/Male ratio was 50/95. Concomitant and adjuvant temozolomide was delivered in 58 (40%) patients. PreRT KPS score was ≤ 70 in 43 (29.7%) and >70 in 102 (70.3%) patients. Sx type was total in 109 (75.2%) patients, subtotal in 36 (24.8%) patients. The interval between Sx and start of RT was median 28 days (range 6-118 days). Percentage of interval groups ≤ 2 weeks, 2-3 weeks, 3-4 weeks, 4-5 weeks and >5 weeks were 8.3%, 24.8%, 25.5%, 13.8%, 27.6%, respectively. For the five groups median survival times were 9, 11, 12, 10 and 16 months, respectively. Overall survival was significantly better for the >5 weeks group compared to 2-3 weeks ($p = 0.021$), 3-4 weeks ($p = 0.008$) and 4-5 weeks groups ($p = 0.02$). Total surgical resection ($p = 0.001$), addition of temozolomide ($p = 0.001$) and Sx-RT interval >5 weeks ($p = 0.036$) were found to be independent prognostic factors for better overall survival by multivariate analysis.

Conclusions: Delay in the initiation of RT is not associated with shorter survival time in patients with GBM. Better survival for patients with longer Sx-RT interval coincides with the RTOG database findings. To explain the radiobiological basis of this unexpected result, further experimental studies and clinical studies in wider populations are needed.

Author Disclosure: M. Abacioglu, None; Z. Akgun, None; A. Ucuncu Kefeli, None; B.M. Atasoy, None; H.B. Caglar, None.

2107 Visual Acuity Outcomes with Palladium 103 (Pd-103) vs. Iodine 125 (I-125) Plaque Treatment for Choroidal Melanoma: The Emory University Experience

B. E. Terakedis, C. Bergstrom, M. Lynn, J. Cleveland, T. M. Aaberg, T. Aaberg, E. Butker, C. Perkins, I. Crocker
Emory University, Atlanta, GA

Purpose/Objective(s): Brachytherapy with I-125 has resulted in good local tumor control rates but with a significant incidence of permanent late visual loss due to radiation optic neuropathy or retinopathy. Pd-103 is a lower energy gamma emitting radioisotope than I-125 and as a result may potentially limit damage to normal ocular structures. Based on these considerations, since 2004 nearly all patients at Emory University have been treated with Pd-103. Most tumors in contact with the optic nerve were treated with I-125 due to documented local control in this high risk population. We report our two year results comparing Pd-103 with I-125.

Materials/Methods: Following IRB approval, we conducted a review of 208 patients treated with radioisotope eye plaques between 7/1993 and 12/2006; complete data was available on 196 patients. Information regarding treatment details, visual acuity, complications of treatment, treatment for local failure [TTT or enucleation], distant metastatic rate, and survival was recorded. The treatment groups were compared using the log-rank test for time to event factors and the Wilcoxon rank sum test for visual acuity.

Results: One hundred fifty-six patients were treated with I-125 and thirty-nine patients treated with Pd-103. There was no difference between the two groups with respect to demographics, tumor location [anterior v. equator v. posterior] or baseline visual acuity. Tumors treated with I-125 (mean lwh = $10.4 \times 10.2 \times 4.2$ mm) were slightly larger than those treated with Pd-103 (mean lwh = $9.3 \times 10.1 \times 2.9$ mm) ($p = 0.02, 0.74$ and <0.01 respectively). Notched plaques were used in 17% of I-125 patients and 5% of Pd-103 patients ($p = .06$). Median follow-up was 3.6 years in the I-125 group and 2.0 years in the Pd-103 group. Local control, distant metastatic disease, and survival rates at 2 years were 95%, 2%, and 96% for the Iodine patients and 100%, 7%, and 94% for the Pd patients. None of these differences were statistically significant. Post-treatment visual acuity appeared better at one year for the Pd-103 patients relative to the I-125 patients ($p = 0.06$) and was significantly better at 2 years ($p = 0.02$). Excluding patients with notched plaques, the difference in post-treatment visual acuity was not significant at one year ($p = .17$) but was borderline significant at 2 years ($p = .06$).